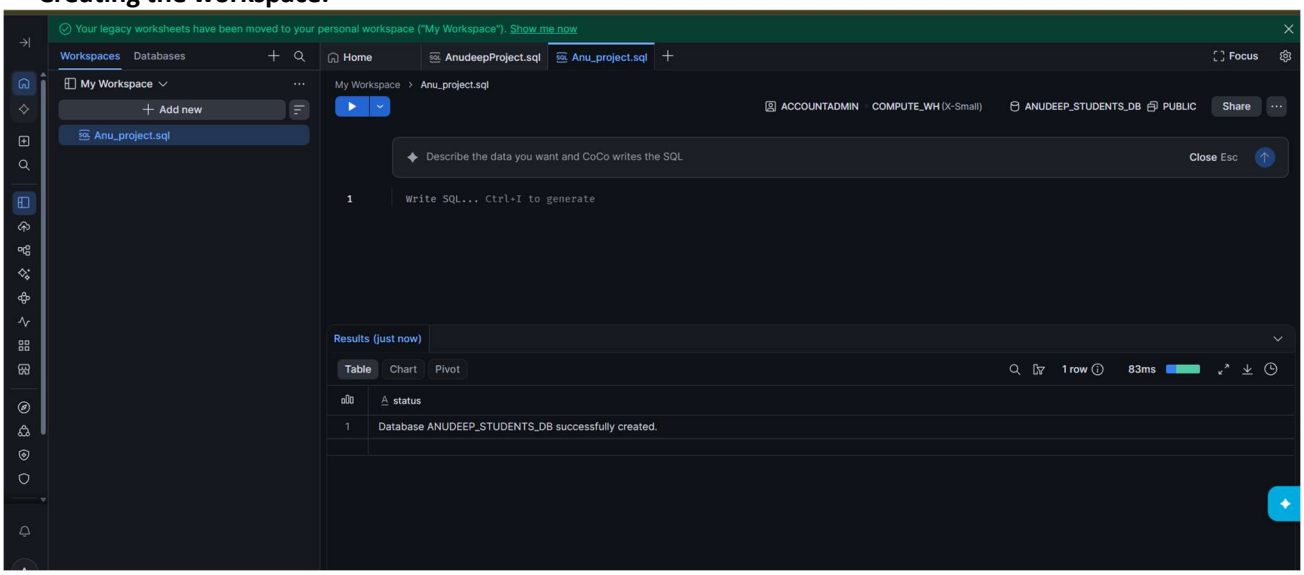
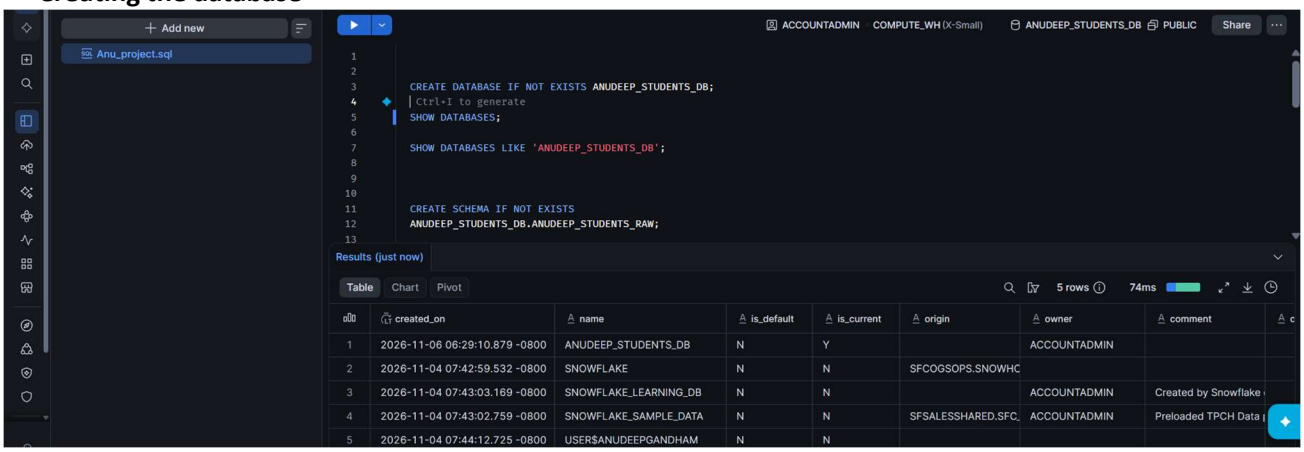
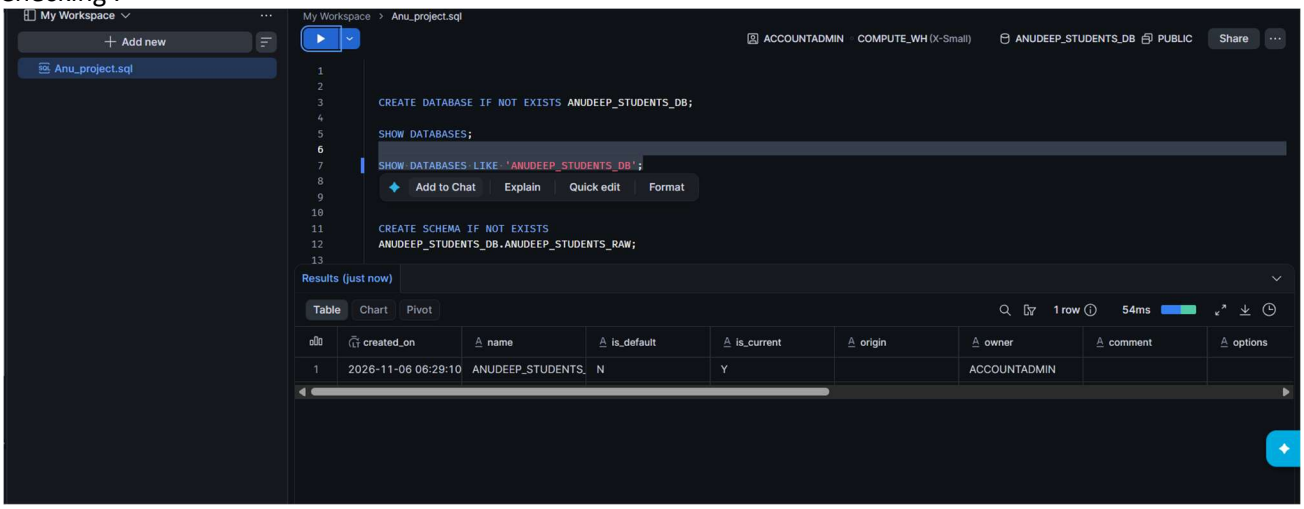
Creating the workspace:



Creating the database



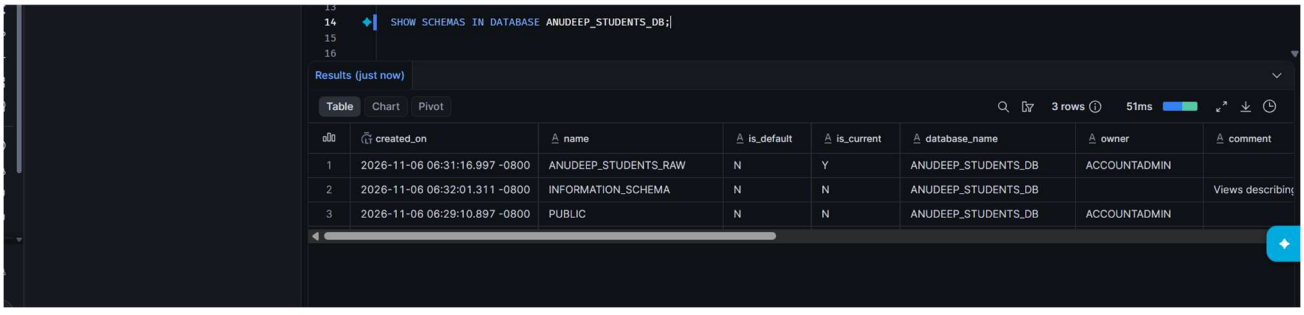
Checking :



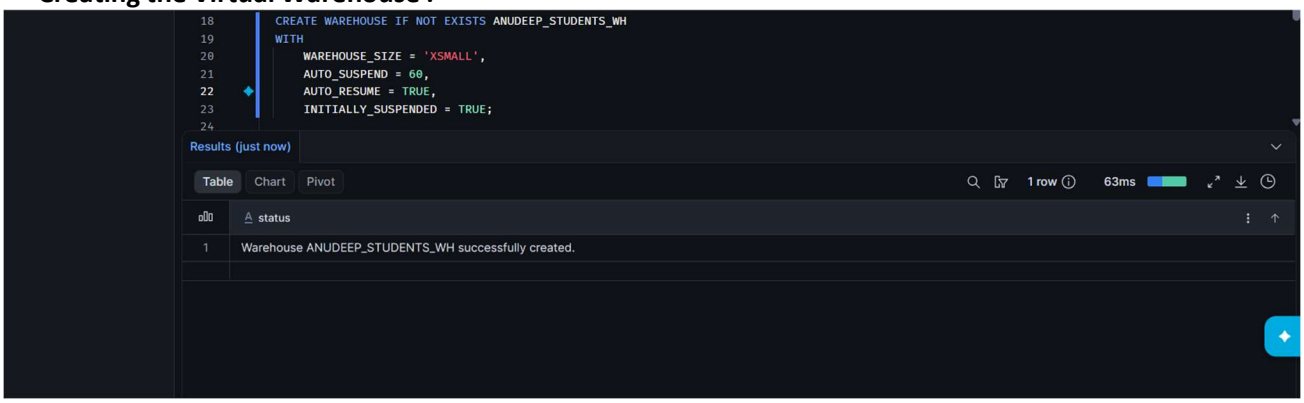
Creating the Schema :



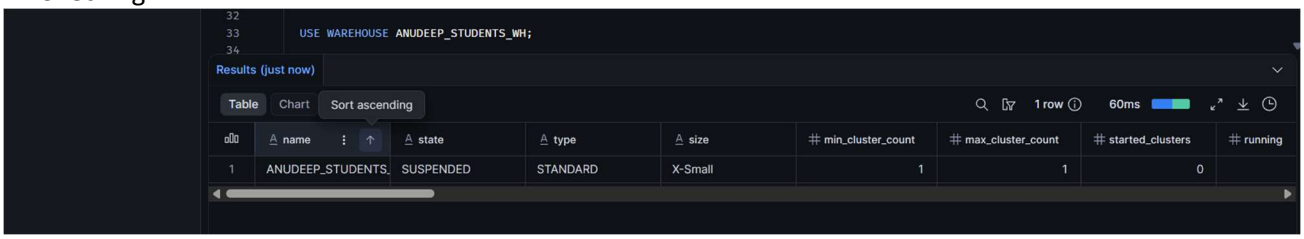
Checking :



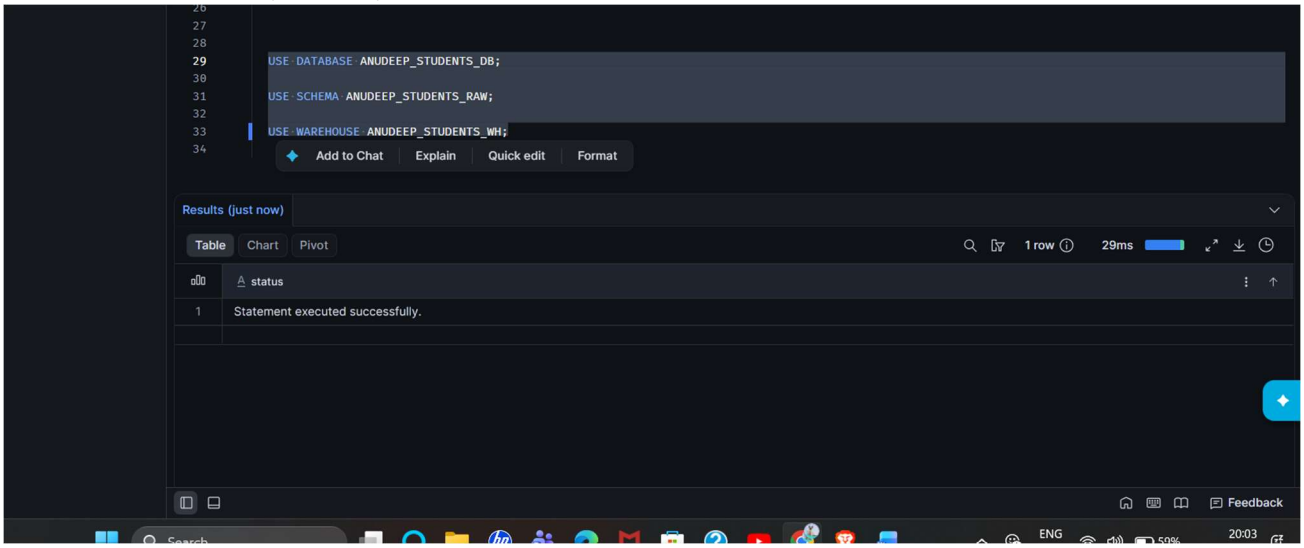
Creating the Virtual Warehouse :



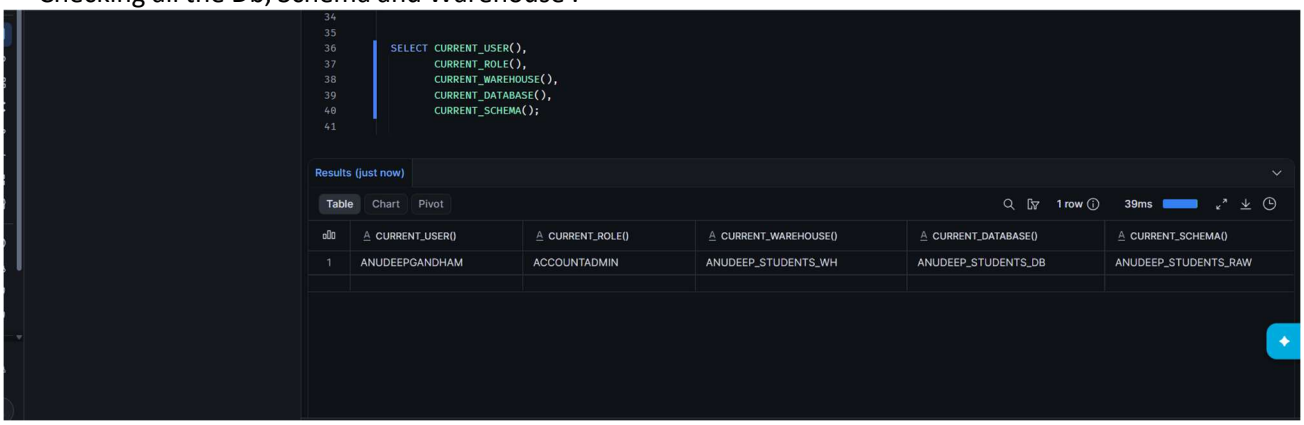
Checking :



Select our database, schema, and warehouse:



Checking all the Db, Schema and Warehouse :



Create the tables:

Students RAW table



Courses RAW table

```sql
CREATE TABLE ANUDEEP_COURSES_RAW_TABLE (
    COURSE_ID VARCHAR(50),
    COURSE_NAME VARCHAR(150),
    CATEGORY VARCHAR(100),
    COURSE_FEE VARCHAR(50),
    DURATION_MONTHS VARCHAR(20)
);
```

Ctrl+I to generate

Results (just now)

Table | Chart | Pivot — 1 row — 160ms

	status
1	Table ANUDEEP_COURSES_RAW_TABLE successfully created.

Enrollments RAW table

```sql
CREATE TABLE ANUDEEP_ENROLLMENTS_RAW_TABLE (
    ENROLLMENT_ID VARCHAR(50),
    STUDENT_ID VARCHAR(50),
    COURSE_ID VARCHAR(50),
    ENROLLMENT_DATE VARCHAR(50),
    ENROLLMENT_STATUS VARCHAR(50)
);
```

Results (just now)

Table | Chart | Pivot — 1 row — 142ms

	status
1	Table ANUDEEP_ENROLLMENTS_RAW_TABLE successfully created.

Checking :

```sql
SHOW TABLES;
```

Results (just now)

Table | Chart | Pivot — 6 rows — 52ms

	created_on	name	database_name	schema_name	kind	comment
1	2026-11-06 06:38:32.160 -0800	ANUDEEP_ATTENDANCE_RAW_TABLE	ANUDEEP_STUDENTS_DB	ANUDEEP_STUDENTS_RAW	TABLE	
2	2026-11-06 06:36:11.532 -0800	ANUDEEP_COURSES_RAW_TABLE	ANUDEEP_STUDENTS_DB	ANUDEEP_STUDENTS_RAW	TABLE	
3	2026-11-06 06:36:40.409 -0800	ANUDEEP_ENROLLMENTS_RAW_TABLE	ANUDEEP_STUDENTS_DB	ANUDEEP_STUDENTS_RAW	TABLE	
4	2026-11-06 06:38:13.905 -0800	ANUDEEP_FACULTY_RAW_TABLE	ANUDEEP_STUDENTS_DB	ANUDEEP_STUDENTS_RAW	TABLE	
5	2026-11-06 06:37:52.923 -0800	ANUDEEP_MARKS_RAW_TABLE	ANUDEEP_STUDENTS_DB	ANUDEEP_STUDENTS_RAW	TABLE	
6	2026-11-06 06:35:31.727 -0800	ANUDEEP_STUDENTS_RAW_TABLE	ANUDEEP_STUDENTS_DB	ANUDEEP_STUDENTS_RAW	TABLE	

Create the CSV file format

```
99
100  CREATE FILE FORMAT IF NOT EXISTS ANUDEEP_STUDENTS_CSV
101  TYPE = CSV
102  FIELD_DELIMITER = ','
103  SKIP_HEADER = 1
104  FIELD_OPTIONALLY_ENCLOSED_BY = '"'
105  EMPTY_FIELD_AS_NULL = TRUE
106  NULL_IF = ('NULL', 'null', '');
107
108
109
110
```

Results (just now)

Table | Chart | Pivot — 1 row — 57ms

	status
1	File format ANUDEEP_STUDENTS_CSV successfully created.

Checking :

```
107
108  SHOW FILE FORMATS;
109
110
```

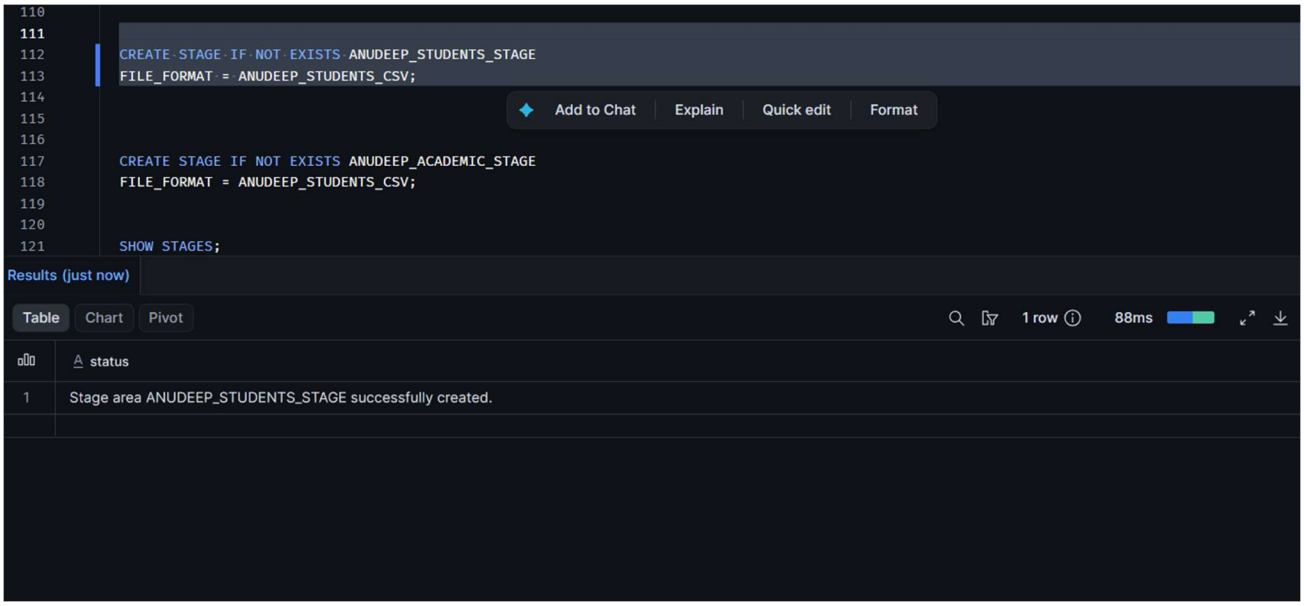
(just now) ··· ×

Table | Chart | Pivot — 1 row — 35ms

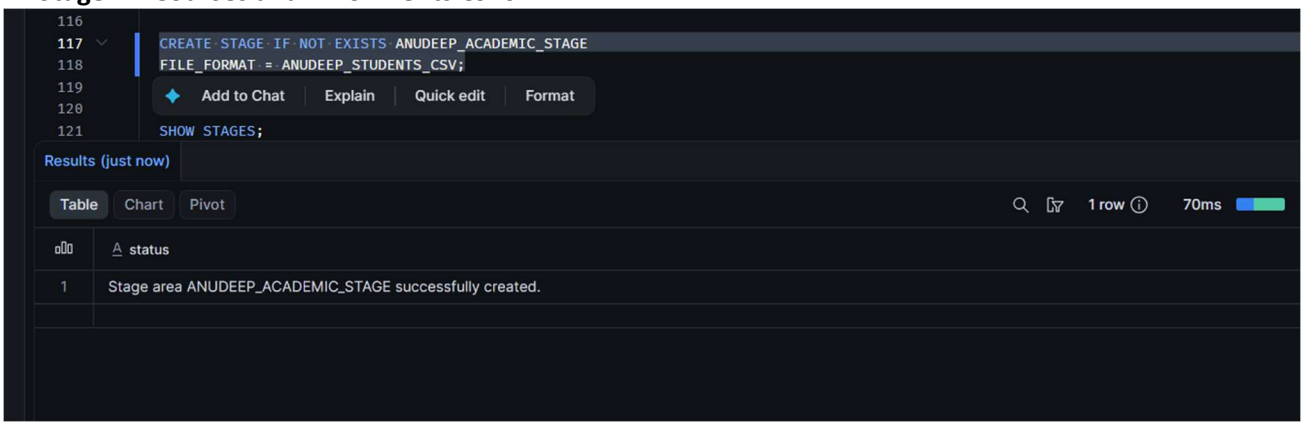
	created_on	name	database_name	schema_name	type	owner	comment	format_optio
1	2026-11-06 06:39:59	ANUDEEP_STUDENTS_	ANUDEEP_STUDENTS_	ANUDEEP_STUDENTS_	CSV	ACCOUNTADMIN		{"TYPE":"CSV",

Create two separate stages

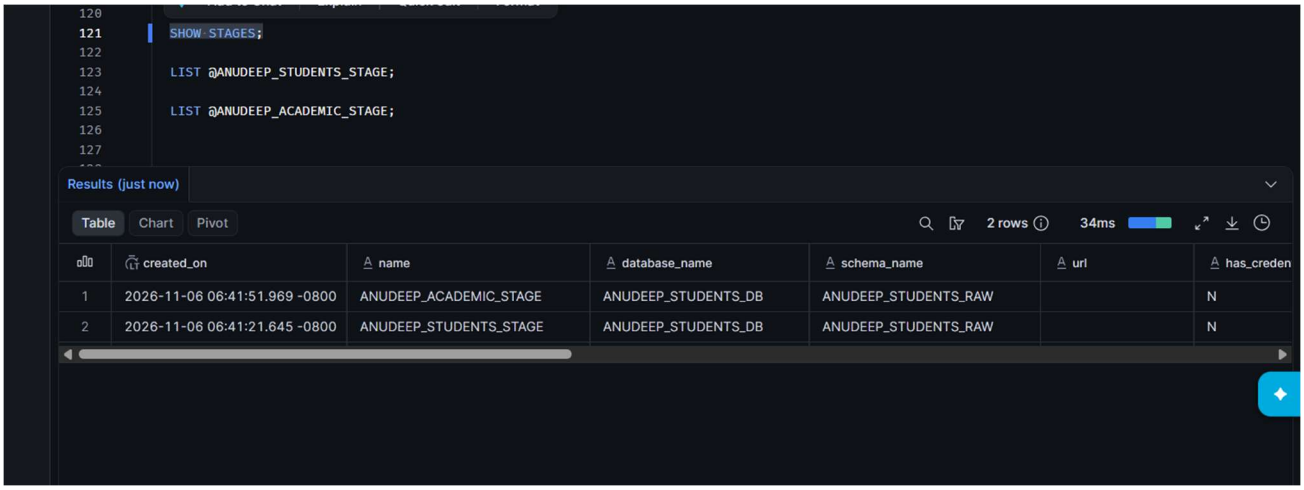
Stage 1 – Students CSV



Stage 2 – Courses and Enrollments CSVs

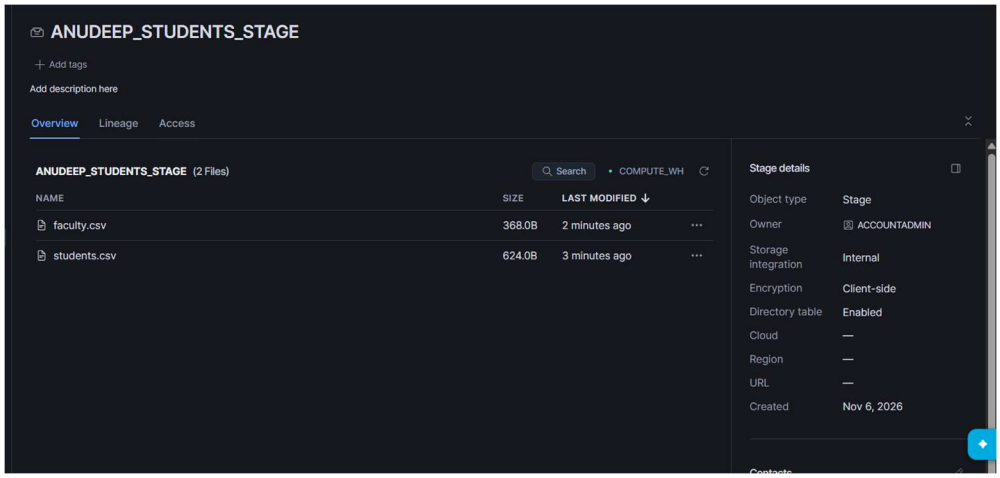


CHECKING:

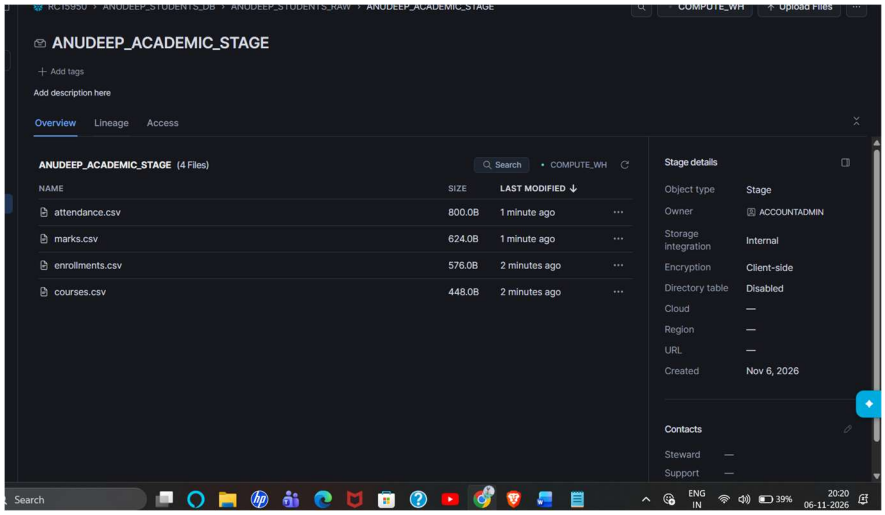


Upload CSV files to the stages

On Students Stage :



Uploading the Multiple Files in the Same stage :



Checking :

```
LIST @ANUDEEP_STUDENTS_STAGE;
```

Results (just now)

Table | Chart | Pivot — 2 rows — 38ms

	name	size	md5	last_modified
1	anudeep_students_stage/faculty.csv	368	0a33290fbb32e947fcde4bb0dbaa3959	Fri, 6 Nov 2026 14:48:20 GMT
2	anudeep_students_stage/students.csv	624	08752ad6d2a4fef2813265ae02b43678	Fri, 6 Nov 2026 14:48:05 GMT

```
LIST @ANUDEEP_ACADEMIC_STAGE;

Ctrl+I to generate
```

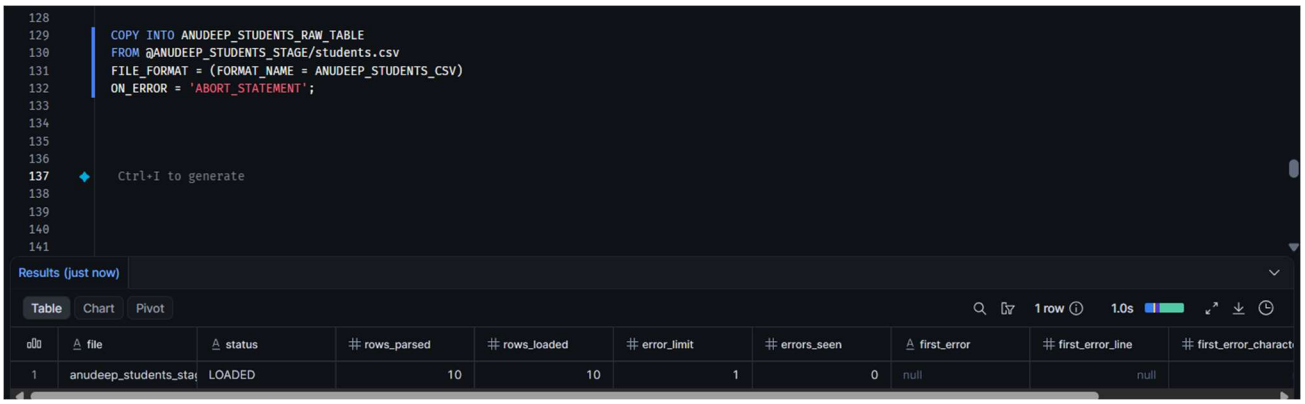
Results (just now)

Table | Chart | Pivot — 4 rows — 45ms

	name	size	md5	last_modified
1	anudeep_academic_stage/attendance.csv	800	4c09f2902cfc6dc603cc788dcc151681	Fri, 6 Nov 2026 14:49:09 GMT
2	anudeep_academic_stage/courses.csv	448	025deebcce7337b803950820a3ad8945	Fri, 6 Nov 2026 14:48:41 GMT
3	anudeep_academic_stage/enrollments.csv	576	61a17f3c1a1f7a291937daec26df3b6f	Fri, 6 Nov 2026 14:48:51 GMT
4	anudeep_academic_stage/marks.csv	624	e031b2f76b7c857328a29d58c6890559	Fri, 6 Nov 2026 14:49:00 GMT

Load the CSV files into their respective RAW tables

Load students



```
COPY INTO ANUDEEP_STUDENTS_RAW_TABLE
FROM @ANUDEEP_STUDENTS_STAGE/students.csv
FILE_FORMAT = (FORMAT_NAME = ANUDEEP_STUDENTS_CSV)
ON_ERROR = 'ABORT_STATEMENT';
```

Ctrl+I to generate

Results (just now)

Table | Chart | Pivot — 1 row — 1.0s

	file	status	rows_parsed	rows_loaded	error_limit	errors_seen	first_error	first_error_line	first_error_charact
1	anudeep_students_sta	LOADED	10	10	1	0	null	null	

Load faculty



```
COPY INTO ANUDEEP_FACULTY_RAW_TABLE
FROM @ANUDEEP_STUDENTS_STAGE/faculty.csv
FILE_FORMAT = (FORMAT_NAME = ANUDEEP_STUDENTS_CSV)
ON_ERROR = 'ABORT_STATEMENT';
```

Ctrl+I to generate

Results (just now)

Table | Chart | Pivot — 1 row — 380ms

	file	status	rows_parsed	rows_loaded	error_limit	errors_seen	first_error	first_error_line	first_error_charact
1	anudeep_students_sta	LOADED	10	10	1	0	null	null	

Load Academic stage info



```
-- LOAD ATTENDANCE DATA

COPY INTO ANUDEEP_ATTENDANCE_RAW_TABLE
FROM @ANUDEEP_ACADEMIC_STAGE/attendance.csv
FILE_FORMAT = (FORMAT_NAME = ANUDEEP_STUDENTS_CSV)
ON_ERROR = 'ABORT_STATEMENT';
```

Results (just now)

Table | Chart | Pivot — 1 row — 375ms

	file	status	rows_parsed	rows_loaded	error_limit	errors_seen	first_error	first_error_line	first_error_charact
1	anudeep_academic_sta	LOADED	20	20	1	0	null	null	

Checking the Loaded Tables:

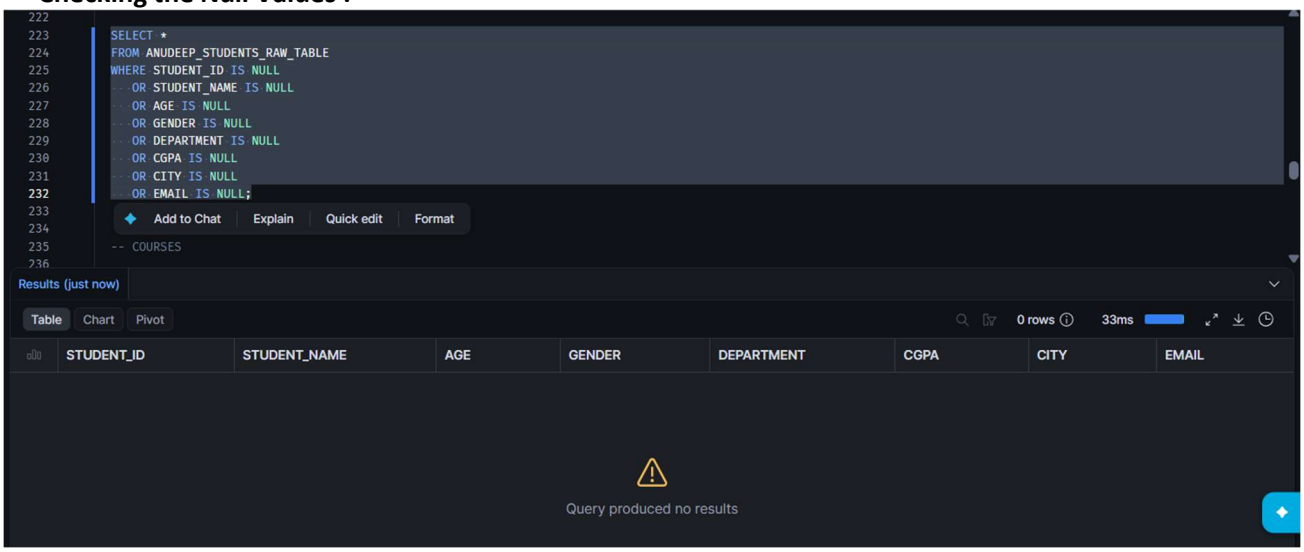


(just now) ⋯ ×

Table | Chart | Pivot — 6 rows — 543ms

	TABLE_NAME	TOTAL_RECORDS
1	STUDENTS	10
2	COURSES	10
3	ENROLLMENTS	15
4	MARKS	20
5	FACULTY	
6	ATTENDANCE	

Checking the Null Values :



```
222
223  SELECT *
224  FROM ANUDEEP_STUDENTS_RAW_TABLE
225  WHERE STUDENT_ID IS NULL
226     OR STUDENT_NAME IS NULL
227     OR AGE IS NULL
228     OR GENDER IS NULL
229     OR DEPARTMENT IS NULL
230     OR CGPA IS NULL
231     OR CITY IS NULL
232     OR EMAIL IS NULL;
233
234
235  -- COURSES
236
```

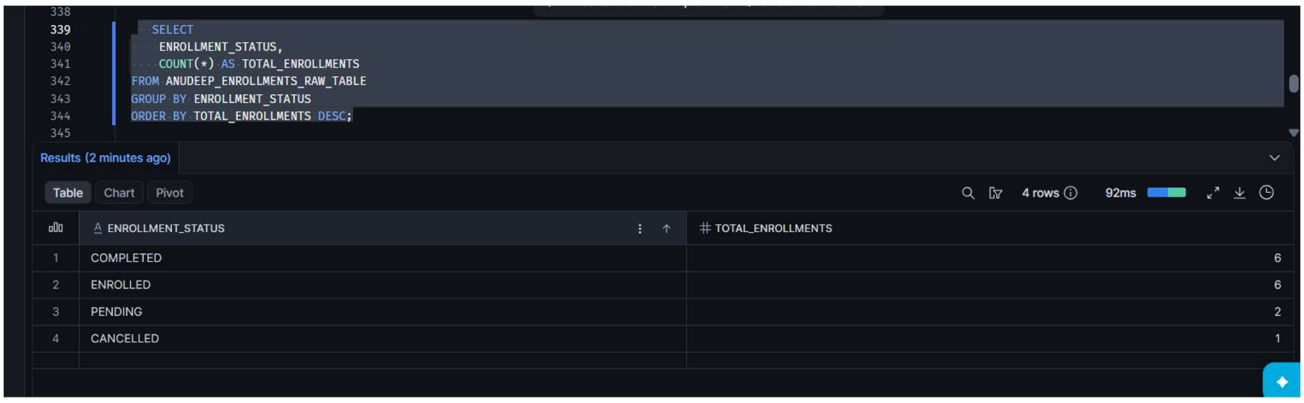
Add to Chat | Explain | Quick edit | Format

Results (just now)

Table | Chart | Pivot — 0 rows — 33ms

STUDENT_ID	STUDENT_NAME	AGE	GENDER	DEPARTMENT	CGPA	CITY	EMAIL

Query produced no results



```
338
339  SELECT
340     ENROLLMENT_STATUS,
341     COUNT(*) AS TOTAL_ENROLLMENTS
342  FROM ANUDEEP_ENROLLMENTS_RAW_TABLE
343  GROUP BY ENROLLMENT_STATUS
344  ORDER BY TOTAL_ENROLLMENTS DESC;
345
```

Results (2 minutes ago)

Table | Chart | Pivot — 4 rows — 92ms

	ENROLLMENT_STATUS	TOTAL_ENROLLMENTS
1	COMPLETED	6
2	ENROLLED	6
3	PENDING	2
4	CANCELLED	1

NULLIF() Function:

Syntax :

NULLIF(expression1, expression2)

It works like this:

- If both expressions are equal, it returns NULL.
- If both expressions are different, it returns the first expression.
- Step 1: `TRIM(STUDENT_NAME)`

Original value	After TRIM
Anudeep	Anudeep
Arjun	Arjun
''	''
' '	''
NULL	NULL

- Step 2: `NULLIF(TRIM(STUDENT_NAME), '')`

Value after TRIM	NULLIF output
Anudeep	Anudeep
Arjun	Arjun
''	NULL
''	NULL

Number Transforming Functions:

TRY_TO_NUMBER() Converts text into numbers safely.

Converts a text value into a number. If conversion is not possible, it returns NULL instead of throwing an error.

Ex:

```
SELECT
  TRY_TO_NUMBER('21') AS AGE1,
  TRY_TO_NUMBER('25') AS AGE2,
  TRY_TO_NUMBER('Twenty') AS AGE3;
```

AGE1 AGE2 AGE3

21 25 NULL

TRY_TO_DECIMAL() Converts CGPA into decimal format safely.

Converts a value into a decimal number while allowing you to specify precision and scale.

Syntax:

TRY_TO_DECIMAL(expression, precision, scale)

Two important concepts:

- Precision: Maximum total number of digits.
- Scale: Maximum number of digits after the decimal point.

```
SELECT
  TRY_TO_DECIMAL('9.75', 5, 2) AS CGPA1,
  TRY_TO_DECIMAL('8.50', 5, 2) AS CGPA2,
  TRY_TO_DECIMAL('Excellent', 5, 2) AS CGPA3;
```

Output:

CGPA1	**CGPA2**	**CGPA3**
9.75	8.50	NULL

Transform Students

```sql
CREATE OR REPLACE TABLE ANUDEEP_STUDENTS AS
SELECT
    TRY_TO_NUMBER(TRIM(STUDENT_ID)) AS STUDENT_ID,

    COALESCE(
        NULLIF(UPPER(TRIM(STUDENT_NAME)), ''),
        'UNKNOWN'
    ) AS STUDENT_NAME,

    TRY_TO_NUMBER(NULLIF(TRIM(AGE), '')) AS AGE,

    COALESCE(
        NULLIF(UPPER(TRIM(GENDER)), ''),
        'UNKNOWN'
    ) AS GENDER,
```

Results (just now)

Table | Chart | Pivot

	status
1	Table ANUDEEP_STUDENTS successfully created.

Transform Enrollments

```sql
CREATE OR REPLACE TABLE ANUDEEP_ENROLLMENTS AS
SELECT
    TRY_TO_NUMBER(TRIM(ENROLLMENT_ID)) AS ENROLLMENT_ID,

    TRY_TO_NUMBER(TRIM(STUDENT_ID)) AS STUDENT_ID,

    TRY_TO_NUMBER(TRIM(COURSE_ID)) AS COURSE_ID,

    TRY_TO_DATE(
        NULLIF(TRIM(ENROLLMENT_DATE), '')
```

Results (just now)

Table | Chart | Pivot | 1 row | 422ms

	status
1	Table ANUDEEP_ENROLLMENTS successfully created.

Transform Marks :

```sql
CREATE OR REPLACE TABLE ANUDEEP_MARKS AS
SELECT
    TRY_TO_NUMBER(TRIM(MARK_ID)) AS MARK_ID,

    TRY_TO_NUMBER(TRIM(STUDENT_ID)) AS STUDENT_ID,

    COALESCE(
        NULLIF(UPPER(TRIM(SUBJECT_NAME)), ''),
        'UNKNOWN'
    ) AS SUBJECT_NAME,

    TRY_TO_NUMBER(TRIM(SEMESTER)) AS SEMESTER,

    TRY_TO_NUMBER(TRIM(MAX_MARKS)) AS MAX_MARKS,

    TRY_TO_NUMBER(TRIM(OBTAINED_MARKS)) AS OBTAINED_MARKS
```

Results (just now)

Table | Chart | Pivot | 1 row | 673ms

	status
1	Table ANUDEEP_MARKS successfully created.

Transform Courses

```
CREATE OR REPLACE TABLE ANUDEEP_COURSES AS
SELECT
    TRY_TO_NUMBER(TRIM(COURSE_ID)) AS COURSE_ID,

    COALESCE(
        NULLIF(UPPER(TRIM(COURSE_NAME)), ''),
        'UNKNOWN'
    ) AS COURSE_NAME,

    COALESCE(
        NULLIF(UPPER(TRIM(CATEGORY)), ''),
```

Results (just now)

Table | Chart | Pivot — 1 row — 392ms

	status
1	Table ANUDEEP_COURSES successfully created.

Transform Faculty

```
CREATE OR REPLACE TABLE ANUDEEP_FACULTY AS
SELECT
    TRY_TO_NUMBER(TRIM(FACULTY_ID)) AS FACULTY_ID,

    COALESCE(
        NULLIF(UPPER(TRIM(FACULTY_NAME)), ''),
        'UNKNOWN'
    ) AS FACULTY_NAME,

    COALESCE(
        NULLIF(UPPER(TRIM(DEPARTMENT)), '')
```

Results (just now)

Table | Chart | Pivot — 1 row — 886ms

	status
1	Table ANUDEEP_FACULTY successfully created.

Transform Attendance:

```
CREATE OR REPLACE TABLE ANUDEEP_ATTENDANCE AS
SELECT
    TRY_TO_NUMBER(TRIM(ATTENDANCE_ID)) AS ATTENDANCE_ID,

    TRY_TO_NUMBER(TRIM(STUDENT_ID)) AS STUDENT_ID,

    COALESCE(
        NULLIF(UPPER(TRIM(SUBJECT_NAME)), ''),
        'UNKNOWN'
    ) AS SUBJECT_NAME,

    TRY_TO_NUMBER(TRIM(TOTAL_CLASSES)) AS TOTAL_CLASSES,

    TRY_TO_NUMBER(TRIM(ATTENDED_CLASSES)) AS ATTENDED_CLASSES,

    TRY_TO_DATE(
        NULLIF(TRIM(ATTENDANCE_DATE), '')
    ) AS ATTENDANCE_DATE
```

Results (just now)

Table | Chart | Pivot — 1 row — 485ms

	status
1	Table ANUDEEP_ATTENDANCE successfully created.

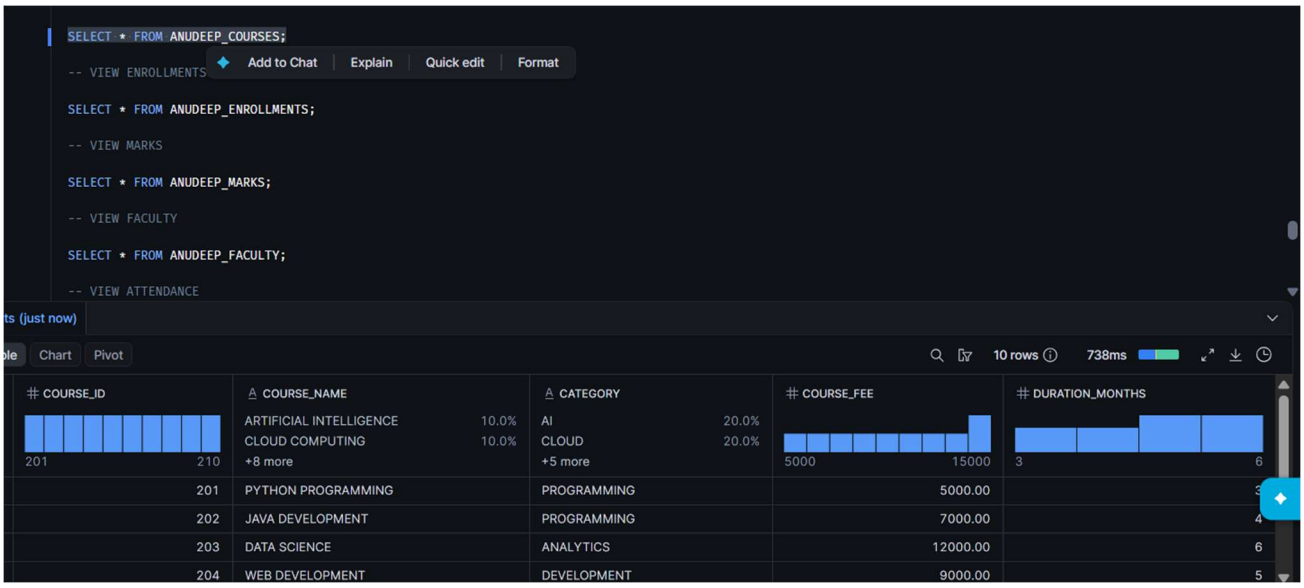
Checking :

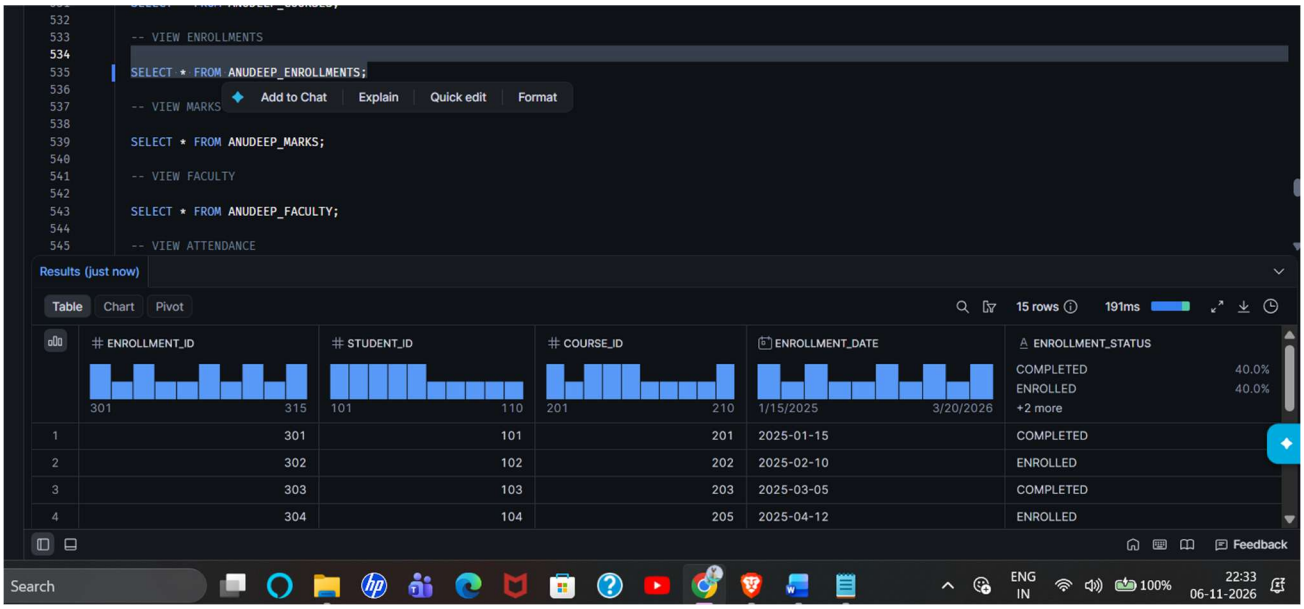
```
524
525     -- VIEW STUDENTS
526
527     SELECT * FROM ANUDEEP_STUDENTS;
528
529     -- VIEW COURSES
530
```

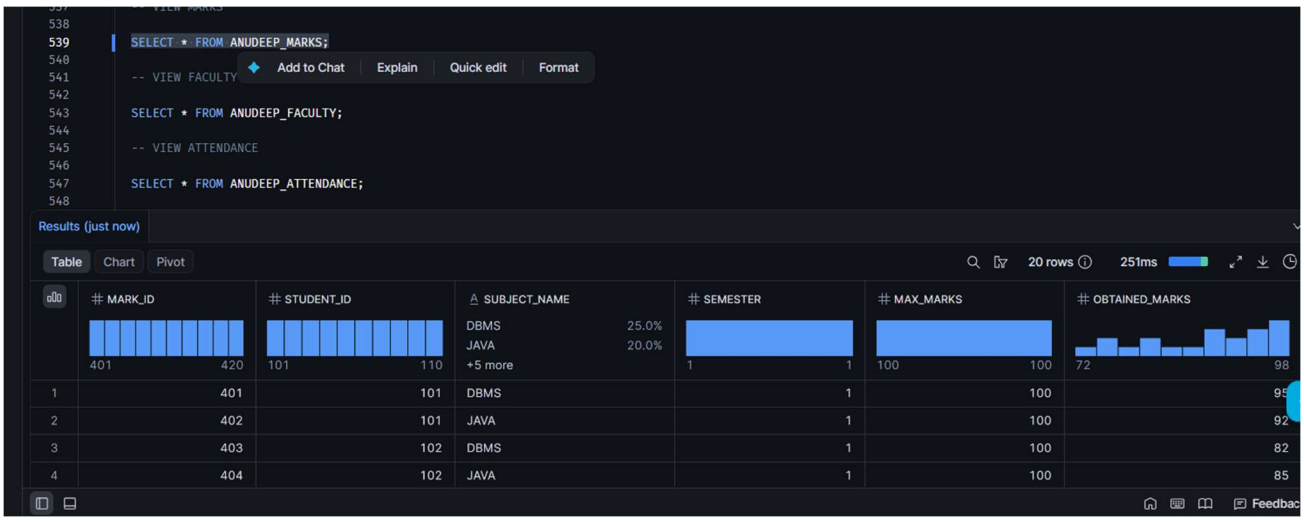
Results (1 hour ago)

Table | Chart | Pivot — 10 rows — 118ms

	STUDENT_ID	STUDENT_NAME	AGE	GENDER	DEPARTMENT	CGPA	CITY	EMAIL
	101 – 110	ANANYA 10.0% ANUDEEP 10.0% +8 more	20 – 23	FEMALE 50.0% MALE 50.0%	CSE 30.0% IT 30.0% +2 more	7.85 – 9.6	HYDERABAD 30.0% CHENNAI 20.0% +3 more	ananya@gmail.com 10.0% anudeep@gmail.com 10.0% +8 more
1	101	ANUDEEP	21	MALE	CSE	9.31	HYDERABAD	anudeep@gmail.com
2	102	RAHUL	22	MALE	ECE	8.75	WARANGAL	rahul@gmail.com
3	103	ANANYA	20	FEMALE	IT	9.12	HYDERABAD	ananya@gmail.com
4	104	SNEHA	21	FEMALE	AIML	8.95	KHAMMAM	sneha@gmail.com


SELECT * FROM ANUDEEP_COURSES;
Add to Chat
Explain
Quick edit
Format
-- VIEW ENROLLMENTS
SELECT * FROM ANUDEEP_ENROLLMENTS;
-- VIEW MARKS
SELECT * FROM ANUDEEP_MARKS;
-- VIEW FACULTY
SELECT * FROM ANUDEEP_FACULTY;
-- VIEW ATTENDANCE
Chart
Pivot
10 rows
738ms
COURSE_ID
COURSE_NAME
CATEGORY
COURSE_FEE
DURATION_MONTHS
201 PYTHON PROGRAMMING PROGRAMMING 5000.00
202 JAVA DEVELOPMENT PROGRAMMING 7000.00
203 DATA SCIENCE ANALYTICS 12000.00
204 WEB DEVELOPMENT DEVELOPMENT 9000.00


-- VIEW ENROLLMENTS
SELECT * FROM ANUDEEP_ENROLLMENTS;
Add to Chat
Explain
Quick edit
Format
-- VIEW MARKS
SELECT * FROM ANUDEEP_MARKS;
-- VIEW FACULTY
SELECT * FROM ANUDEEP_FACULTY;
-- VIEW ATTENDANCE
Results (just now)
Table
Chart
Pivot
15 rows
191ms
ENROLLMENT_ID
STUDENT_ID
COURSE_ID
ENROLLMENT_DATE
ENROLLMENT_STATUS
1 301 101 201 2025-01-15 COMPLETED
2 302 102 202 2025-02-10 ENROLLED
3 303 103 203 2025-03-05 COMPLETED
4 304 104 205 2025-04-12 ENROLLED
Feedback
Search
ENG IN
100%
22:33
06-11-2026


SELECT * FROM ANUDEEP_MARKS;
Add to Chat
Explain
Quick edit
Format
-- VIEW FACULTY
SELECT * FROM ANUDEEP_FACULTY;
-- VIEW ATTENDANCE
SELECT * FROM ANUDEEP_ATTENDANCE;
Results (just now)
Table
Chart
Pivot
20 rows
251ms
MARK_ID
STUDENT_ID
SUBJECT_NAME
SEMESTER
MAX_MARKS
OBTAINED_MARKS
1 401 101 DBMS 1 100 95
2 402 101 JAVA 1 100 92
3 403 102 DBMS 1 100 82
4 404 102 JAVA 1 100 85

```
-- VIEW FACULTY

SELECT * FROM ANUDEEP_FACULTY;

-- VIEW ATTENDANCE
```

Add to Chat | Explain | Quick edit | Format

sults (just now)

Table | Chart | Pivot — 10 rows — 157ms

	FACULTY_ID	FACULTY_NAME	DEPARTMENT	EXPERIENCE_YEARS	CITY
	501 – 510	DR KUMAR 10.0% DR LAKSHMI 10.0% +8 more	CSE 30.0% IT 30.0% +2 more	6 – 18	HYDERABAD 30.0% CHENNAI 20.0% +3 more
1	501	DR KUMAR	CSE	15	HYDERABAD
2	502	DR RAMESH	ECE	12	WARANGAL
3	503	MS PRIYA	IT	8	HYDERABAD
4	504	MR SURESH	AIML	10	KHAMMAM

```
-- VIEW ATTENDANCE

SELECT * FROM ANUDEEP_ATTENDANCE;
```

Add to Chat | Explain | Quick edit | Format

Results (just now)

Table | Chart | Pivot — 20 rows — 185ms

	ATTENDANCE_ID	STUDENT_ID	SUBJECT_NAME	TOTAL_CLASSES	ATTENDED_CLASSES	ATTENDANCE_DATE
	601 – 620	101 – 110	DBMS 25.0% JAVA 20.0% +5 more	50 – 50	34 – 50	1/31/2026 – 1/31/2026
1	601	101	DBMS	50	48	2026-01-31
2	602	101	JAVA	50	45	2026-01-31
3	603	102	DBMS	50	40	2026-01-31
4	604	102	JAVA	50	42	2026-01-31

```
DESC TABLE ANUDEEP_STUDENTS;
```

Add to Chat | Explain | Quick edit | Format

esults (just now)

Table | Chart | Pivot — 8 rows — 40ms

	name	type	kind	null?	default	primary key	unique key	check	expression	comment
1	STUDENT_ID	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null	null
2	STUDENT_NAME	VARCHAR(300)	COLUMN	Y	null	N	N	null	null	null
3	AGE	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null	null
4	GENDER	VARCHAR(60)	COLUMN	Y	null	N	N	null	null	null
5	DEPARTMENT	VARCHAR(300)	COLUMN	Y	null	N	N	null	null	null
6	CGPA	NUMBER(4,2)	COLUMN	Y	null	N	N	null	null	null

```
DESC TABLE ANUDEEP_ATTENDANCE;
```

Add to Chat | Explain | Quick edit | Format

Results (just now)

Table | Chart | Pivot — 6 rows — 26ms

	name	type	kind	null?	default	primary key	unique key	check	expression	comment
1	ATTENDANCE_ID	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null	null
2	STUDENT_ID	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null	null
3	SUBJECT_NAME	VARCHAR(300)	COLUMN	Y	null	N	N	null	null	null
4	TOTAL_CLASSES	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null	null
5	ATTENDED_CLASSES	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null	null
6	ATTENDANCE_DATE	DATE	COLUMN	Y	null	N	N	null	null	null

Perform analytical queries

Find students belonging to CSE department

```
SELECT
    STUDENT_ID,
    STUDENT_NAME,
    DEPARTMENT,
    CGPA
FROM ANUDEEP_STUDENTS
WHERE DEPARTMENT = 'CSE';
```

Results (just now)

Table | Chart | Pivot — 3 rows — 373ms

	STUDENT_ID	STUDENT_NAME	DEPARTMENT	CGPA
1	101	ANUDEEP	CSE	9.3
2	105	ARJUN	CSE	7.8
3	109	VARUN	CSE	7.0

Display unique departments and cities

Add to Chat | Explain | Quick edit | Format

```
SELECT DISTINCT DEPARTMENT
FROM ANUDEEP_STUDENTS;
```

Results (just now)

Table | Chart | Pivot — 4 rows — 26ms

	DEPARTMENT
1	CSE
2	ECE
3	IT
4	AIML

```
SELECT DISTINCT CITY
FROM ANUDEEP_STUDENTS;
```

Results (just now)

Table | Chart | Pivot

	CITY
1	HYDERABAD
2	WARANGAL
3	KHAMMAM
4	CHENNAI
5	VIJAYAWADA

Find students whose age is between 20 and 22

```
570  SELECT *
571  FROM ANUDEEP_STUDENTS
572  WHERE AGE BETWEEN 20 AND 22
573  ORDER BY AGE ASC;
```

Results (just now)

Table | Chart | Pivot — 8 rows — 374ms

	STUDENT_ID	STUDENT_NAME	AGE	GENDER	DEPARTMENT	CGPA	CITY	EMAIL
1	103	ANANYA	20	FEMALE	IT	9.12	HYDERABAD	ananya@gmail.com
2	106	PRIYA	20	FEMALE	ECE	9.45	VIJAYAWADA	priya@gmail.com
3	110	DIVYA	20	FEMALE	IT	8.85	KHAMMAM	divya@gmail.com
4	101	ANUDEEP	21	MALE	CSE	9.31	HYDERABAD	anudeep@gmail.com
5	104	SNEHA	21	FEMALE	AIML	8.95	KHAMMAM	sneha@gmail.com
6	108	MEGHANA	21	FEMALE	AIML	9.60	HYDERABAD	meghana@gmail.com

Calculate total students and average CGPA

```
575
576  SELECT
577      COUNT(*) AS TOTAL_STUDENTS,
578      ROUND(AVG(CGPA), 2) AS AVERAGE_CGPA,
579      MAX(CGPA) AS HIGHEST_CGPA,
580      MIN(CGPA) AS LOWEST_CGPA
581  FROM ANUDEEP_STUDENTS;
582
583
584      Ctrl+I to generate
585
586
```

Results (just now)

Table | Chart | Pivot — 1 row — 73ms

	TOTAL_STUDENTS	AVERAGE_CGPA	HIGHEST_CGPA	LOWEST_CGPA
1	10	8.80	9.60	7.85

Perform JOIN operations

Display student enrollment and course details

```
583
584  SELECT
585      S.STUDENT_ID,
586      S.STUDENT_NAME,
587      S.DEPARTMENT,
588      C.COURSE_NAME,
589      C.CATEGORY,
590      E.ENROLLMENT_DATE,
591      E.ENROLLMENT_STATUS
592  FROM ANUDEEP_STUDENTS S
593  INNER JOIN ANUDEEP_ENROLLMENTS E
594      ON S.STUDENT_ID = E.STUDENT_ID
595  INNER JOIN ANUDEEP_COURSES C
596      ON E.COURSE_ID = C.COURSE_ID
597  ORDER BY S.STUDENT_ID;
598
```

Results (just now)

Table | Chart | Pivot — 15 rows — 818ms

	STUDENT_ID	STUDENT_NAME	DEPARTMENT	COURSE_NAME	CATEGORY	ENROLLMENT_DATE	ENROLLMENT_STATUS
	101 – 110	ANANYA 13.3%, ANUDEEP 13.3%, +8 more	CSE 33.3%, IT 26.7%, +2 more	DATA SCIENCE 13.3%, DEVOPS ENGINEERING 13.3%, +8 more	AI 20.0%, CLOUD 20.0%, +5 more	1/15/2025 – 3/20/2026	COMPLETED 40.0%, ENROLLED 40.0%, +2 more
1	101	ANUDEEP	CSE	PYTHON PROGRAMMING	PROGRAMMING	2025-01-15	COMPLETED
2	101	ANUDEEP	CSE	DATA SCIENCE	ANALYTICS	2025-11-12	ENROLLED
3	102	RAHUL	ECE	JAVA DEVELOPMENT	PROGRAMMING	2025-02-10	ENROLLED
4	102	RAHUL	ECE	WEB DEVELOPMENT	DEVELOPMENT	2025-12-01	COMPLETED

Calculate student-wise average marks

```
599
600  SELECT
601      S.STUDENT_ID,
602      S.STUDENT_NAME,
603      S.DEPARTMENT,
604      COUNT(M.MARK_ID) AS TOTAL_SUBJECTS,
605      ROUND(AVG(M.OBTAINED_MARKS), 2) AS AVERAGE_MARKS,
606      SUM(M.OBTAINED_MARKS) AS TOTAL_MARKS
607  FROM ANUDEEP_STUDENTS S
608  INNER JOIN ANUDEEP_MARKS M
609      ON S.STUDENT_ID = M.STUDENT_ID
610  GROUP BY
611      S.STUDENT_ID,
612      S.STUDENT_NAME,
613      S.DEPARTMENT
614  ORDER BY AVERAGE_MARKS DESC;
```

Results (just now)

Table | Chart | Pivot — 10 rows — 273ms

	STUDENT_ID	STUDENT_NAME	DEPARTMENT	TOTAL_SUBJECTS	AVERAGE_MARKS	TOTAL_MARKS
	101 – 110	ANANYA 10.0%, ANUDEEP 10.0%, +8 more	CSE 30.0%, IT 30.0%, +2 more	2 – 2	74 – 96.5	148 – 193
1	108	MEGHANA	AIML	2	96.50	193
2	106	PRIYA	ECE	2	96.00	192
3	101	ANUDEEP	CSE	2	93.50	187
4	103	ANANYA	IT	2	93.50	187

Calculate student-wise average marks using AVG() OVER()



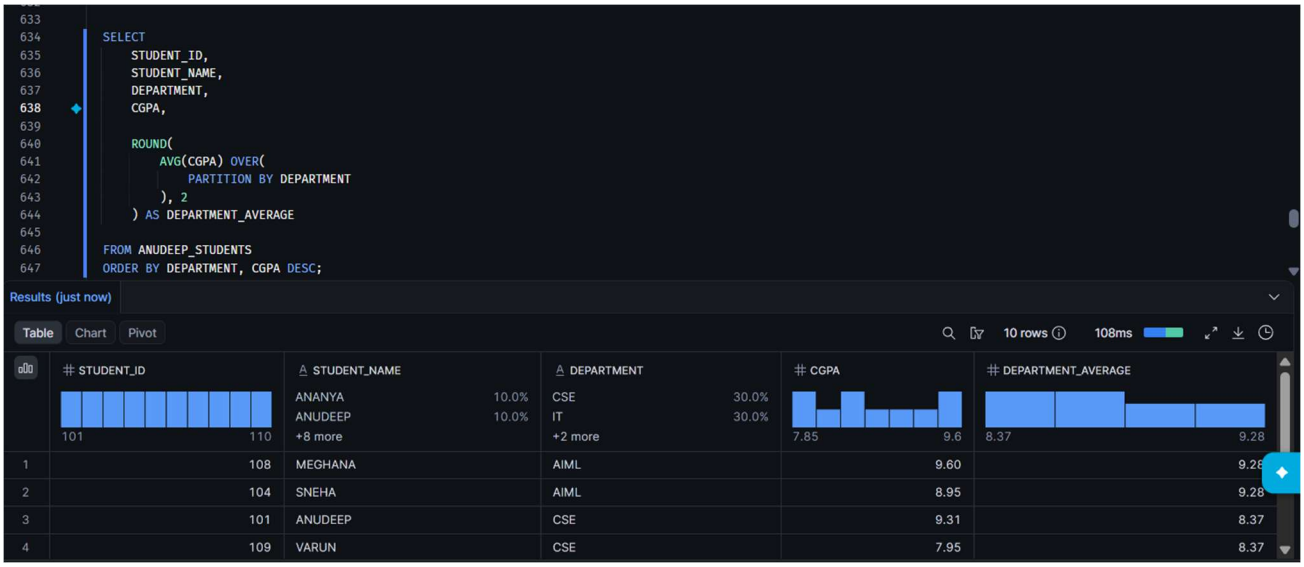
```
617
618  SELECT
619      STUDENT_ID,
620      SUBJECT_NAME,
621      OBTAINED_MARKS,
622
623      ROUND(
624          AVG(OBTAINED_MARKS) OVER(
625              PARTITION BY STUDENT_ID
626          ), 2
627      ) AS STUDENT_AVERAGE
628
629  FROM ANUDEEP_MARKS
630  ORDER BY STUDENT_ID;
631
```

Results (just now)

Table | Chart | Pivot — 20 rows — 318ms

	STUDENT_ID	SUBJECT_NAME	OBTAINED_MARKS	STUDENT_AVERAGE
	101 – 110	DBMS 25.0%, JAVA 20.0%, +5 more	72 – 98	74 – 96.5
1	101	DBMS	95	93.50
2	101	JAVA	92	93.50
3	102	DBMS	82	83.50
4	102	JAVA	85	83.50

Calculate department-wise average CGPA



```
633
634  SELECT
635      STUDENT_ID,
636      STUDENT_NAME,
637      DEPARTMENT,
638      CGPA,
639
640      ROUND(
641          AVG(CGPA) OVER(
642              PARTITION BY DEPARTMENT
643          ), 2
644      ) AS DEPARTMENT_AVERAGE
645
646  FROM ANUDEEP_STUDENTS
647  ORDER BY DEPARTMENT, CGPA DESC;
```

Results (just now)

Table | Chart | Pivot — 10 rows — 108ms

	STUDENT_ID	STUDENT_NAME	DEPARTMENT	CGPA	DEPARTMENT_AVERAGE
	101 – 110	ANANYA 10.0%, ANUDEEP 10.0%, +8 more	CSE 30.0%, IT 30.0%, +2 more	7.85 – 9.6	8.37 – 9.28
1	108	MEGHANA	AIML	9.60	9.28
2	104	SNEHA	AIML	8.95	9.28
3	101	ANUDEEP	CSE	9.31	8.37
4	109	VARUN	CSE	7.95	8.37